

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MID SEMESTER EXAMINATION MO23)

CLASS: BTECH / IMSC
BRANCH: BT/CIVIL/CHEMICAL/MECH/PIE/FT/PHYSICS

SEMESTER : I
SESSION : MO2023

SUBJECT: CS101 PROGRAMMING FOR PROBLEM SOLVING
TIME: 02 Hours

FULL MARKS: 25

INSTRUCTIONS:

1. The question paper contains 5 questions each of 5 marks and total 25 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Tables/Data handbook/Graph paper etc., if applicable, will be supplied to the candidates

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|---|-----|----|-----|
| Q.1(a) Explain any two examples of each of the following operators.
a) Arithmetic operators
b) Logical operators | [2] | 2 | BL1 |
| Q.1(b) Explain briefly about each line of code in the following program.
#include<stdio.h>
void main()
{
int a=10;
float b= 23.56;
char c='A';
printf("%d %f %c\n",a,b,c);
} | [3] | 1 | BL2 |
| Q.2(a) Make a flow chart to find out whether a number is an odd number or even number, if any integer number is input through keyboard. | [2] | 1 | BL2 |
| Q.2(b) Write all steps to solve expression involving integer values only which is based on precedence and associativity concept.
5*(10/5+ (7*6-3)% (6+5-2)) | [3] | 2 | BL2 |
| Q.3(a) Write any four common flowchart symbols along with their meaning. | [2] | 1 | BL1 |
| Q.3(b) Write a program to calculate the sum of the digits of a 3-digit input number. | [3] | 2 | BL3 |
| Q.4(a) Find the output of the following program
#include <stdio.h>
void main()
{
int i=0, x=0;
for (i=1;i<5;++i)
{
if (i%2==1)
x+=i;
else
x--;
printf("%d ", x);
}
} | [2] | 2 | BL3 |
| Q.4(b) Write a program to determine whether an entered character is a vowel or not using switch case statement. | [3] | 2 | BL2 |
| Q.5(a) Write the equivalent code using while loop in place of for loop
int i, sum=0;;
for(i=1; i<=5; i++)
{
sum=sum+i;
} | [2] | 2 | BL2 |
| Q5(b) Write an interactive program that will read in a positive integer value and determine whether the integer is a prime number or not.
Example: 2, 3, 5, 7, 11 are prime numbers. | [3] | 2 | BL2 |